

Experimental Investigation of the Steel-Concrete Joint in a Hybrid Tower for a Wind Turbine under Fatigue Loading

Moon Ok Kim*, Tae Min Kim**, Dae Yong Lee***, and Moon Kyum Kim****

Received June 10, 2018/Revised January 29, 2019/Accepted March 12, 2019/Published Online April 30, 2019

Abstract

This study investigated the fatigue behaviors at the joint of a hybrid steel–concrete wind turbine tower. A fatigue design approach was proposed for the joint based on CEB-FIP model code 1990 and Eurocode 2. In addition, to estimate the long-term fatigue performance related to continuous wind changes at the joint, two types of specimens were fabricated and tested. Each specimen consisted of steel and concrete sections, which were connected with anchor bolts, and they were designed to investigate the effect of the embedded length of the anchor bolt. Two million cycles of cyclic loading were applied to the specimens, after which a monotonic static loading was applied to investigate the residual performance of each specimen. The experimental results demonstrate that the joint of a hybrid steel–concrete wind turbine tower designed using the proposed approach retained sufficient fatigue performance. Thus, this approach can be effective for fatigue design of the joint.

Keywords: *hybrid wind turbine tower, steel-concrete joint, anchor bolt, fatigue test, onshore tower*

1. Introduction

Demand for renewable energy technologies has continuously increased across the globe to cope with depletion of fossil fuels and environmental pollution. Of several renewable energy sources, wind power generation has received a large share of the attention, and the size of wind power systems has been increasing to enhance their economic efficiency and to use winds of good quality (GWEC, 2016). However, large wind power systems cause problems related to the transportation of construction members; for a steel wind turbine tower, the transported fabricated structures can exceed bridge clearance heights (Von der Harr and Marx, 2015). To overcome these transportation issues, several types of wind turbine towers, which are made of concrete materials and can be constructed on-site, have been proposed (Alvarez-Anton *et al.*, 2016; Staffan *et al.*, 2010). A hybrid-type tower can provide advantages by using precast concrete and structural steel to make the lower and upper parts of the wind turbine tower, respectively. This allows the transportation difficulties due to the large size of the tower to be solved by on-site assembly. In addition, the construction efficiency and applicability to various sites can be enhanced by using standard tubular steel segments.

The fabrication cost of a wind turbine tower is roughly 26% that of the total system including the wind turbine (EWEA,

2009). Furthermore, damage to the tower can negatively affect operation of the wind turbine system. This leads to serious economic costs, and thus these damages should be prevented during the design process. In the design of a wind turbine tower, safety checks are conducted with respect to the ultimate strength as well as the fatigue resistance of the materials; fatigue loads are a major factor in the design of the tower stiffness because the tower is exposed to continuous vibration from the wind turbine (Bai *et al.*, 2017).

For hybrid wind turbine towers, it is necessary to investigate the fatigue failure of precast concrete and steel members as well as the concrete–steel joints. The fatigue safety of the first two members can be addressed using conventional design codes. However, fatigue criteria for the concrete–steel joint member in the design codes is not clear. In addition, there has been little research on fatigue at the joint; there have been studies on the joint between steel members (Petersen, 1998; Schmidt and Neuper, 1997), but not the concrete–steel joint. Although some experiments have been conducted on the anchor bolt used to connect steel and concrete parts (Rodriguez *et al.*, 2001; Delhomme *et al.*, 2010, 2016; Hariyadi *et al.*, 2017), these were pull-out tests of the anchor bolts intended to analyze the static or dynamic behavior of the joint. Studies on fatigue in anchor bolts are not extensive, and those focused on the joint in wind turbine towers are particularly lacking. This study presents an approach to

*Member, General Manager, R&D Laboratory, SPEC Engineering Y&P Co., Ltd., Seoul 05854, Korea (Corresponding Author, E-mail: hihioho@speceng.co.kr)

**Member, Senior Researcher, Infrastructure Safety Research Center, K-water Convergence Institute, Daejeon 34045, Korea (E-mail: taemin@kwater.or.kr)

***Member, Senior Project Manager, Steel Solution Center, POSCO, Korea (E-mail: dragon70@posco.com)

****Member, Professor, Dept. of Civil and Environmental Engineering, Yonsei University, Seoul 03722, Korea (E-mail: applymkk@yonsei.ac.kr)

fatigue design for the steel-concrete joint of wind turbine towers with anchor bolts and demonstrates the validity of the fatigue design through several fatigue experiments. Two types of joint between the steel and precast concrete were used in the specimens. This paper consists of three parts. In the first part, several design codes related to fatigue are compared, and a design approach is proposed by reviewing the codes. Then, the design of steel-concrete joint specimens for fatigue testing is described in detail. Finally, the experimental results from fatigue and monotonic loading tests are analyzed.

2. Review of Design Codes for Fatigue

The steel-concrete joint of a hybrid wind turbine tower is subjected to periodic and dynamic moment forces. These forces

act alternately as compressive and tensile on the joint, and cause fatigue compressive and tensile failures in the concrete part, including the anchor bolts, as shown in Fig. 1. Generally, a joint fatigue analysis can be conducted for the steel and concrete parts as well as the anchor bolt. For steel, the S-N curves in widely used design codes can be used to inform the fatigue safety analysis. The curves are similar to each other, which enables fatigue design results to be consistent. In contrast, fatigue criteria for concrete in the design codes differ, and results of the fatigue safety analyses are thus dependent on which criterion was used. Furthermore, there are no clear criteria for conducting fatigue design analyses for anchor bolts.

2.1 Concrete

The Eurocode 2 (EN 1992-1-1, 2004), GL wind guideline

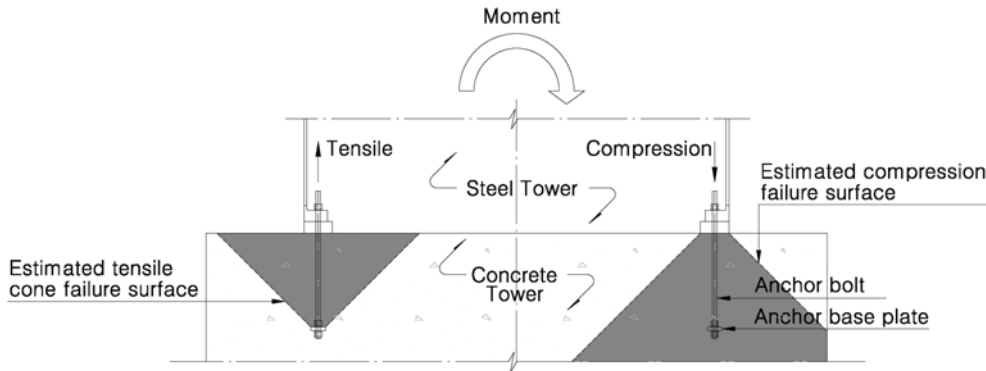


Fig. 1. Failure Behavior of a Steel-Concrete Joint due to Moment Force

Table 1. Equations for Evaluating Fatigue Safety of Concrete from the Eurocode 2, GL Wind Guideline, and CEB-FIP Model Code 1990

Eurocode 2	GL wind guideline	CEB-FIP model code 1990
Under compression: $\frac{\sigma_{c,max}}{f_{cd,fat}} \leq 0.5 + 0.45 \cdot \frac{\sigma_{c,min}}{f_{cd,fat}}$ ≤ 0.9 for $f_{ck} \leq 50$ MPa ≤ 0.8 for $f_{ck} > 50$ MPa	Under compression: $S_{cd,max} \leq 0.4 + 0.46 \cdot S_{cd,min}$ Where, $S_{cd,min} = \gamma_{sd} \cdot \sigma_{c,min} \cdot \eta_c / f_{cd,fat}$ $S_{cd,max} = \gamma_{sd} \cdot \sigma_{c,max} \cdot \eta_c / f_{cd,fat}$	Under pure compression for $0 < S_{c,min} \leq 0.8$: $\log N_1 = (12 + 16S_{c,min} + 8S_{c,min}^2)(1 - S_{c,max})$ $\log N_2 = 0.2 \log N_1 (\log N_1 - 1)$ $\log N_3 = \log N_2 (0.3 - 0.375S_{c,min}) / \Delta S_c$ Under compression-tension with $\sigma_{ct,max} \leq 0.026 \sigma_{c,max} :$ $\log N = 9(1 - S_{c,max})$ Under pure tension and tension-compression with $\sigma_{ct,max} > 0.026 \sigma_{c,max} :$ $\log N = 12(1 - S_{ct,max})$ Where, $S_{c,max} = \sigma_{c,max} / f_{ck,fat}$ $S_{c,min} = \sigma_{c,min} / f_{ck,fat}$ $\Delta S_c = S_{c,max} - S_{c,min} $ $S_{ct,max} = \sigma_{ct,max} / f_{ctk,min}$

Note: $\sigma_{c,max}$ is the maximum compressive stress, $\sigma_{c,min}$ is the minimum compressive stress, $\sigma_{ct,max}$ is the maximum tensile stress, γ_{sd} is the partial safety factor, η_c is the factor for the non-uniform stress distribution, $f_{cd,fat}$ is the design value for the fatigue strength of the concrete under compressive loading, $f_{ck,fat}$ is the fatigue reference compressive strength of the concrete, and $f_{ctk,min}$ is the minimum characteristic tensile strength.

(GL, 2010), and CEB-FIP model code 1990 (CEB, 1993) were compared to establish a fatigue design approach for the concrete part of a hybrid wind turbine tower. Both Eurocode 2 and GL wind guideline estimate the fatigue resistance in concrete using stresses due to compression and shear forces. These codes apply zero stress to the tensile area of concrete to conservatively estimate fatigue safety. Rather than this method, fatigue checks using the stresses and number of cycles at the reinforcement or prestressing steel are recommended. Unlike these codes, CEB-FIP model code 1990 provides S-N curves to evaluate the compression and tensile fatigue in concrete. The concrete fatigue design formulae from the Eurocode 2, GL wind guideline, and CEB-FIP model code 1990 are listed in Table 1.

For a hybrid wind turbine tower, tensile fatigue failures are expected to occur at anchor bolts embedded in concrete, and the fatigue behavior depends strongly on the anchor bolt arrangement in the joint. Thus, a design method with respect to the tensile fatigue of concrete is necessary to consider the effects of the anchor bolt. In this study, the fatigue safety of concrete was evaluated according to CEB-FIP model code 1990; this code allows a fatigue failure area to be roughly and conservatively predicted.

2.2 Anchor Bolts

Because there are no prescribed fatigue performance criteria for anchor bolts at the joint, this study applied a design approach in which steel bolts used in steel structures are considered as the prestressing steel used in prestressed concrete structures. This approach not only considers the similarity of the load transfer mechanism between steel bolts and embedded prestressing steel with respect to post-tension, but it also leads to more conservative design of the anchor bolts. Each method to evaluate the fatigue safety of the steel bolts and the prestressing steel is provided in Eurocode 3 (EN 1993-1-1, 2005) and Eurocode 2 (EN 1992-1-1, 2004), respectively (see Fig. 2). In this study, the anchor bolts were treated as prestressing steel, and a fatigue evaluation of the

post-tensioning single strands and straight/curved tendons in plastic ducts was examined instead of anchor bolts.

3. Design of Specimens for Fatigue Experiments

3.1 Overview of Target Hybrid Tower at Test Bed

In this study, the fatigue performance of a hybrid tower for a 2

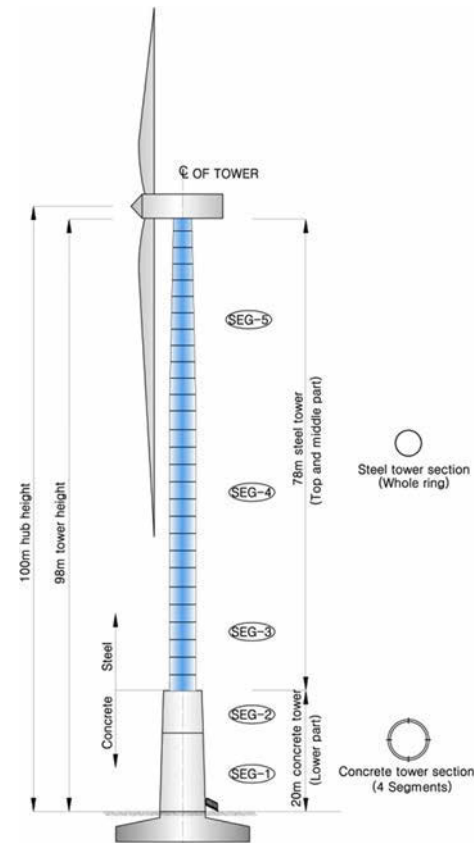


Fig. 3. Schematic Diagram of the Target Hybrid Tower Consisting of Steel and Precast Concrete

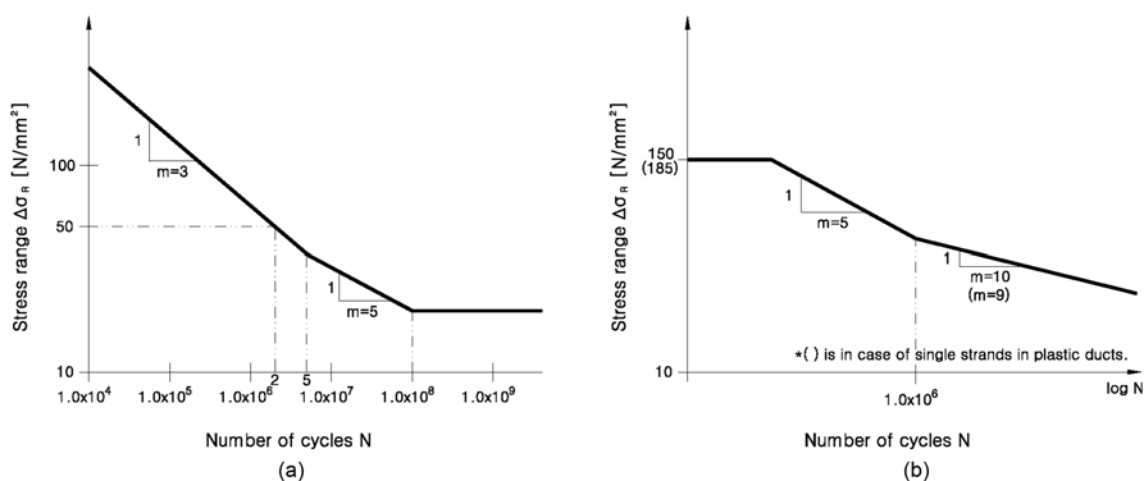


Fig. 2. S-N Curves for a Steel Bolt and Prestressing Steel: (a) Steel Bolt, (b) Post-tensioning Prestressing Steel (Straight Tendons in Plastic Ducts)

Table 2. Major Specifications of the Wind Turbine for the Hybrid Tower Test Bed

Rated power	Hub height	Nacelle mass	Number of rotor blades	Design lifetime
2,000 kW	100.0 m	124,218 kg	3 pcs	20 y

Table 3. Major Loads at the Top and Joint for the Test Bed

Location	Ultimate loads		Fatigue loads*
	Moment (kN·m)	Axial force (kN)	Moment (kN·m)
Top of tower	7,358	1,621	991
Steel-concrete interface	66,094	3,914	4,410

*The moment is calculated at 1.83×10^8 cycles, and the slope of the S-N curve is 3.

MW wind turbine at a test bed was analyzed. This tower consists of steel and precast concrete for the upper (78 m) and lower (20 m) parts of the tower, respectively, which are connected with anchor bolts (see Fig. 3). Both parts were prefabricated and assembled at the construction site; major specifications of the wind turbine are summarized in Table 2. The external load cases applied to the tower were calculated according to the GL wind guideline (GL, 2010). Cross-sectional forces such as an axial force and bending moment at the steel-concrete joint (EL 20 m) and top of the tower (EL 98 m) are listed in Table 3. These forces are equivalent to the ultimate and fatigue limit states, respectively, and are considered as external forces acting on the joint. A cross-section of the joint in the hybrid tower is shown in Fig. 4. The embedded M42 anchor bolts are used to connect the precast concrete part with an L-shape flange attached to the steel part and the anchor bolts are placed 115 mm apart on the top surface of the concrete part.

3.2 Calculation of the Pull-Out Force

For the hybrid wind turbine tower at the test bed, the equivalent

fatigue moment at the joint given in Table 3 was calculated for a wind turbine designed to operate at 0.29 Hz for 20 y. The fatigue performance of materials has been generally estimated for repeated loadings of 1 or 2 million cycles. Because increasing the number of cycles drastically increases the amount of time and cost required for the fatigue tests, the fatigue moment at 183 million cycles ($0.29 \text{ cycles} \times 60 \text{ sec} \times 60 \text{ min} \times 24 \text{ hr} \times 365 \text{ d} \times 20 \text{ y}$) should be converted to that at 2 million cycles. Assuming a linear relationship between the uniaxial bending stress and moment, and using the relationship between the number of cycles and the stress variation to generate the specific fatigue damage of the material, the fatigue moment at 2 million cycles can be estimated from that at 183 million cycles using Eq. (1):

$$\Delta\sigma_{n1} = \left(\frac{N_{n2}}{N_{n1}} \times \Delta\sigma_{n2}^m \right)^{1/m} \quad (1)$$

where $\Delta\sigma_n$ is the stress range at n cycles, N_n is the number of cycles in stress range $\Delta\sigma_n$, and m is the slope of the S-N curve.

The tensile force intensity along the perimetric direction of the joint surface due to bending moments was calculated using Eq. (2), and the resultant tensile force was assumed to be uniformly distributed at each anchor bolt. Based on this assumption, the tensile force acting on a single anchor bolt was calculated by multiplying the tensile intensity by the distance between the anchor bolts:

$$Z = \left\{ \frac{M_{eq}}{\frac{\pi(D_1 - D_2)^4}{64}} \times \frac{D_1}{2} \right\} \times t \quad (2)$$

where D_1 and D_2 are the outer and inner diameter of the steel shell at the joint, respectively; Z is the tensile force intensity; M_{eq} is the equivalent fatigue moment; and t is the thickness of the steel shell at the joint. The tensile forces of each anchor bolt under fatigue loading were calculated using Eqs. (1) and (2), and

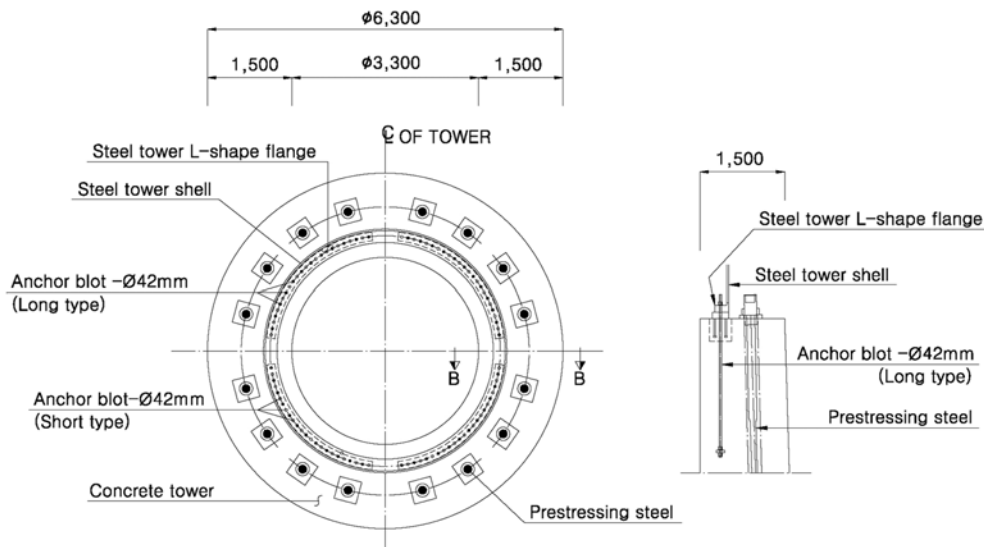


Fig. 4. Cross-Section of the Joint between the Steel and Concrete Parts of the Hybrid Tower Test Bed

Table 4. Calculation of the Tensile Load Acting on an Anchor Bolt due to an Equivalent Fatigue Load

Fatigue life	M_{eq} (kN·m)	Z (kN/mm)	T (kN)
1.83×10^8 cycles	4,410	310.1	43.3
2.00×10^6 cycles	19,872	1,397.4	195.3

are listed in Table 4.

In Table 4, T is the axial force of the anchor bolt. However, during fatigue tests, a cyclic load of 147.1 kN was applied to the specimen, which is 75.3% of the tensile force of 195.3 kN. This load reduction is made to consider decreases in the load resistance capacity of the specimen; while a large number of reinforcement bars were placed in the concrete around the steel-concrete joint in the test bed tower to resist extreme tensile loads, those in the test specimen are much less. The reduction ratio was estimated by assuming that decreases in the fatigue capacity due to reinforcement reduction are proportional to those in the ultimate pull-out resistance. The ratio of the extreme pull-out resistance of the specimen (458.4 kN) to that of the joint at the test bed (608.7 kN) was applied to calculate the decreased fatigue loading; the joint force at the test bed was calculated using extreme external axial and bending forces of 3,914 kN and 66,094 kN-m, respectively. The pull-out resistances were estimated according to the tensile strength formulae presented in the “Korea structural concrete design code and commentary” (KCI, 2012). Examining the possible ultimate failure states and the corresponding resistance forces of the laboratory specimen listed in Table 5, it was expected that concrete failure near the anchor bolt would occur first among these failure modes. Thus, 458.4 kN was selected as the extreme tensile resistance of the specimen. In Table 5, the notation is as follows: N_{sa} is the nominal strength of an anchor in

Table 5. Calculation of the Extreme Tensile Resistance Strength of the Joint

Ultimate state	Equation	Results
Steel strength of anchor	$N_{sa} = A_{se,N} f_{uta}$	963.2 kN
Concrete breakout*	$N_{cb} = \frac{A_{Nc}}{A_{Nco}} \psi_{ed,N} \psi_{c,N} \psi_{cp,N} N_b$	458.4 kN
Concrete pullout	$N_{pn} = \psi_{c,p} N_p$	1,018.8 kN
Concrete side-face blowout	$N_{sb} = (13c_{a1} \sqrt{A_{brg}}) \lambda \sqrt{f_{ck}}$	1,411.5 kN

*When reinforcement bars are installed in the concrete around anchor bolts, specimen failure is governed by the yield of the reinforcement. Thus, concrete failure was evaluated through the yield of the reinforcement.

tension as governed by the steel, N_{cb} is the nominal concrete breakout strength in tension of an anchor, N_{pn} is the nominal pull-out strength in tension of an anchor, N_{sb} is the nominal side-face blowout strength of an anchor, $A_{se,N}$ is the effective cross-sectional area of an anchor, f_{uta} is the specified tensile strength of the anchor steel, A_{Nc} is the projected concrete failure area of an anchor, A_{Nco} is the projected concrete failure area of an anchor not limited by edge distance, $\psi_{ex,N}$ is a factor used to modify the tensile strength of anchors based on proximity to the edges of a concrete member, $\psi_{c,N}$ is a factor used to modify the tensile strength of anchors based on the presence or absence of cracks in the concrete, $\psi_{cp,N}$ is a factor used to modify the tensile strength of post-installed anchors intended for use in uncracked concrete without supplementary reinforcement, N_b is the basic concrete breakout strength in tension of an anchor in cracked concrete, $\psi_{c,P}$ is a factor used to modify the pullout strength of anchors based on the presence or absence of cracks in the concrete, N_p is

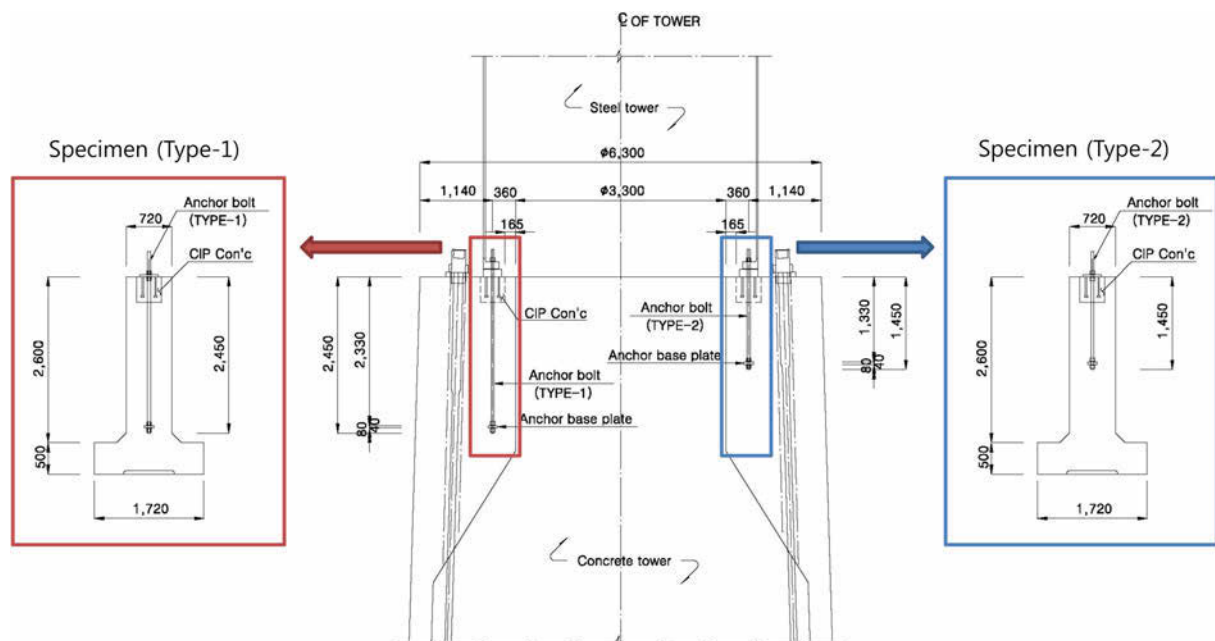


Fig. 5. Detailed Diagram of the Steel-Concrete Joint

the pullout strength in tension of an anchor in cracked concrete, c_{al} is the distance from the center of an anchor shaft to the edge of concrete in one direction, A_{brg} is the bearing area of the head of a stud or anchor bolt, λ is a modification factor related to the unit weight of concrete, and f_{ck} is the specified compressive strength of concrete.

3.3 Test Specimen Specifications

Full-scale specimens were used to analyze fatigue behavior at the concrete–steel joint of the tower. In the real tower joint, long and short anchor bolts were placed alternately to reduce bearing stresses in the concrete parts and prestressing steels were placed to introduce compressive stress into the concrete parts as shown in Fig. 5. Unlike the real tower joint, each specimen had only an anchor bolt embedded in the concrete part; anchor bolts ($\Phi = 42$ mm) with lengths of 1.45 m and 2.45 m were used. In the real tower, post-tensioning force of 4,930 kN was considered for each individual prestressing steel resulting in a compressive stress of 8.9 MPa in concrete member. According to the CEB-FIP model code 1990, however, fatigue of concrete was conservatively considered under pure tension state rather than the compressive state. Therefore, the specimens were fabricated and tested, excluding prestressing steel.

Although embedding one anchor bolt in the specimen made its joint behavior relatively simple and clear, the validity of the test results using these specimens cannot be guaranteed. To verify the effectiveness of using these specimens and determine the dimension of its cross-section, stress analyses were conducted using a finite element model of the specimen. In the finite element model, a pull-out force of 147.1 kN was applied to the top of the anchor bolt, and fixed boundary conditions were imposed on the bottom of the concrete foundation. Steel and concrete were assumed to be linear materials, and their material

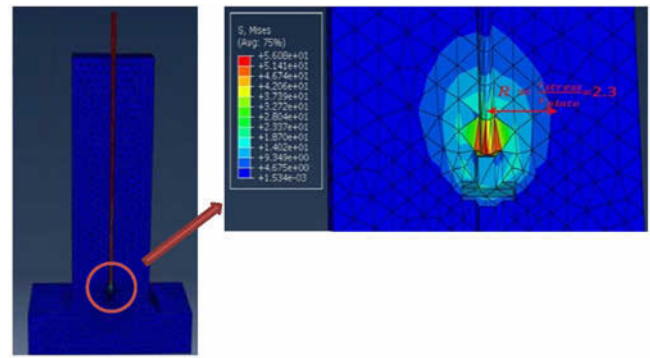


Fig. 6. Stress Distribution near the Anchor Bolt

properties are listed in Table 6. The properties of concrete at 28 d and a high-strength bolt (10.9 grade) in the test bed were applied to the model. From the stress distribution around the anchor base plate under the anchor bolts (see Fig. 6), the cross-sectional dimensions of the joint specimens were determined to be rectangular (720 mm $\times$ 860 mm) by considering the distance from each anchor bolt and the peak stress near the anchor bolts; a drastic increase in the stresses occurred within 345 mm (about twice the width of the anchor base plate) of the anchor bolts.

Based on the above preliminary design of the joint specimen, the specific dimensions and reinforcement placement were determined as shown in Fig. 7, and two types of test specimens with different length anchor bolts were manufactured, as shown in Fig. 8. Before the cyclic and static loading tests, concrete strength tests were conducted. The compressive strength of the concrete satisfied the design requirement of 40 MPa; the average strength of concrete for specimens of Type-1 and Type-2 were 40.7 MPa and 45 MPa, respectively.

Table 6. Material Properties of the Concrete and Anchor Bolt

Concrete				Anchor bolt			
f_{ck} (MPa)	f_{ctk} (MPa)	E_c (MPa)	ν_c	f_{yb} (MPa)	f_{ub} (MPa)	E_b (MPa)	ν_b
40	2.5	30,891	0.18	900	1,000	210,000	0.30

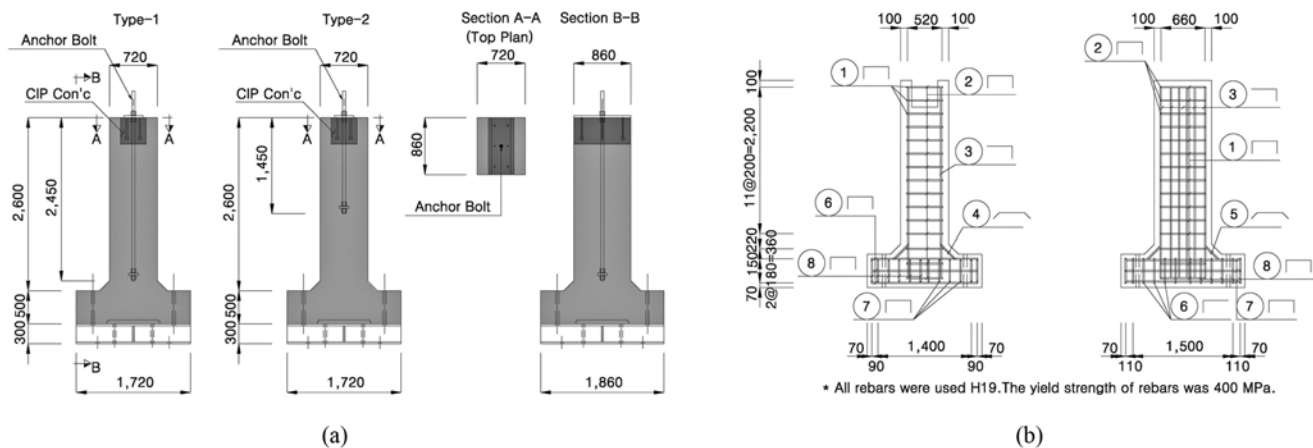


Fig. 7. Schematics of the Test Specimens: (a) Types and Dimensions of Specimens, (b) Reinforcement Placement

3.4 Fatigue Resistance Assessment

3.4.1 Concrete

The fatigue resistance of concrete was estimated based on the CEB-FIP model code 1990 (CEB, 1993). Assuming plain concrete subjected to pure tensile load, the fatigue safety was evaluated using Eq. (3):

$$\log N = 12(1 - S_{ct,max}) \quad (3)$$

where $S_{ct,max}$ is the tensile stress level ($\sigma_{ct,max}/f_{ct,k,min}$), $\sigma_{ct,max}$ is the maximum tensile stress due to the external load, and $f_{ct,k,min}$ is the minimum tensile strength proposed in Eurocode for concrete with a compressive strength of 40 MPa; $\sigma_{ct,max}$ and $f_{ct,k,min}$ are 0.24 MPa and 2.5 MPa, respectively, and thus $S_{ct,max}$ is 0.095. The concrete fatigue resistance was determined by applying $S_{ct,max}$ to the S-N curve, as shown in Fig. 8. As $S_{ct,max}$ is 0.095, which is less than the maximum value of $S_{ct,max}$ at 2 million cycles (0.475), there are a sufficient number of cycles to reach concrete fatigue.

3.4.2 Anchor Bolt

The fatigue resistance of the anchor bolts was calculated using

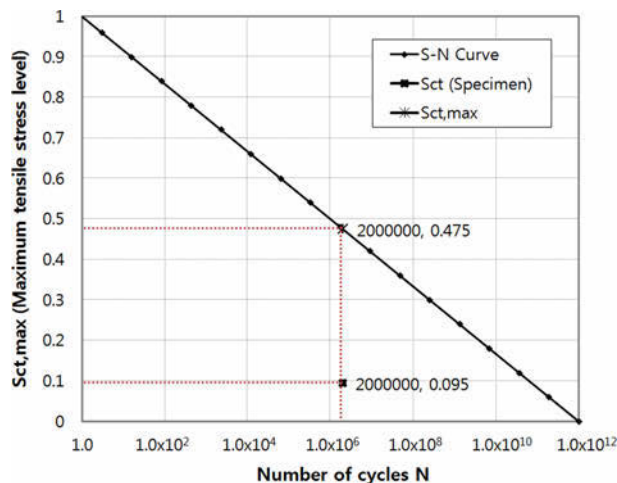


Fig. 8. S-N Curve for Concrete subjected to a Tensile Force

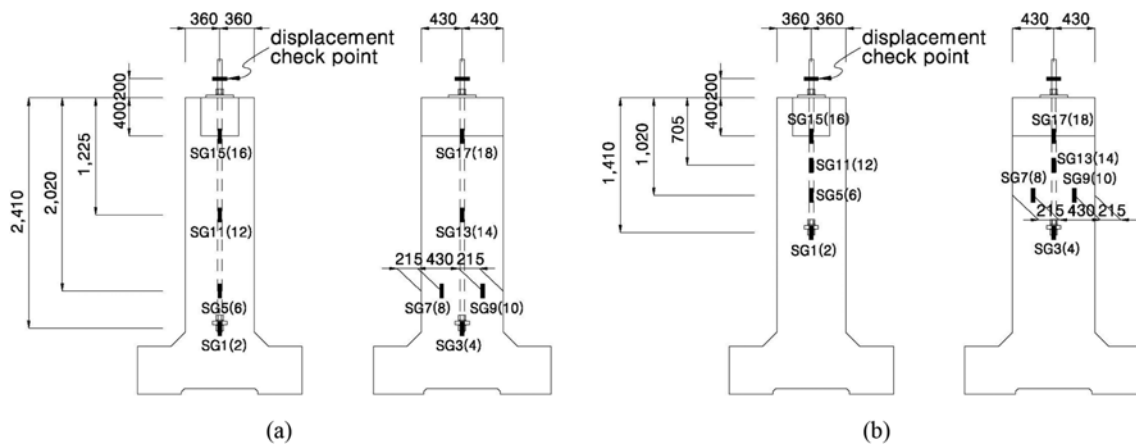


Fig. 9. Measurement Locations: (a) Test Specimen: Type-1, (b) Test Specimen: Type-2

Table 7. Fatigue Stress Ranges of Prestressing Steel

Post-tensioning types	Stress range (MPa)	
	At 1×10^6 cycles	At 2×10^6 cycles
Single strands in plastic ducts	185	171
Straight tendons or curved tendons in plastic ducts	150	140

the fatigue evaluation method of prestressing steel for post-tensioning in Eurocode 2 (EN 1992-1-1, 2004). Because only a stress range at 1 million cycles is presented in Eurocode 2, the stress range was converted to that at 2 million cycles using Eq. (1); cycle slopes of 9 and 10 were adopted for single strands and straight/curved tendons in plastic ducts, respectively. The stress ranges are listed in Table 7. The tensile stress of the anchor bolts due to the test load was calculated to be 106.2 MPa, and thus the fatigue safety is satisfied. The stress was derived by dividing the pull-out force by the effective area of 1,385 mm².

4. Experiments and Results

4.1 Overview of Experiments

Loading tests were conducted to evaluate the fatigue performance of the joint specimen and to verify the proposed fatigue assessment guidelines. The experiment consists of two steps: First, a cyclic loading with 2 million cycles, and second, monotonic and static loading with a pull-out force to investigate the static residual loading capacity of the specimen.

A fatigue testing machine (FTM) with a capacity of 2,000 kN was used to conduct the fatigue tests. To set up the specimen, the head of the anchor bolt and the foundation part of the specimen were fixed to the FTM and the ground, respectively. The cyclic loading range was from 0 – 147.1 kN. The applied load cycle had a frequency of 2 Hz, and the total number of cycles was 2 million. Furthermore, the concrete strain, displacement of the anchor bolt, and crack opening width were measured as the number of cycles increased. Strain gages were installed as shown in Fig. 9 and described in Table 8. The displacement at the head

Table 8. Detailed Description of Strain Gage Locations

Strain gage number	Strain gage location
SG1 – SG4	At lower end of anchor bolt
SG5 – SG10	At 45° away above the lower end of anchor bolt
SG11 – SG14	At 1/2 of the anchor bolt embedment length
SG15 – SG18	On the surface of the CIP concrete and precast concrete

of the anchor bolt was monitored by the FTM. A ruler was used to measure the crack opening width.

4.2 Displacement

Figure 10 shows several load-displacement curves obtained during cyclic loading. As the number of cycles increased, the slope of the curves decreased while the maximum displacement of the test specimen increased. Although the curves show that the specimen stiffness decreased with increasing number of cycles, the maximum displacement changes by less than 1 mm and is relatively gradual. Thus, the specimen is stable with respect to fatigue. It was noted that the displacement increment from the first cycle to 0.5 million cycles is greater than the others. This was caused by detachment of a cast-in-placed (CIP) part from the specimen after one cycle, which resulted in a change in the embedded length of the anchor bolt.

The load-displacement curves for each specimen were compared to analyze the effect of the embedded length of the anchor bolt. At each cycle, the maximum displacement of the Type-1 (longer anchor bolt) specimen is larger than the Type-2 specimen, as is the displacement increment with increasing number of cycles. However, the difference between the displacement in the Type-1 and Type-2 specimens is small because it is due to the tensile deformation that is proportional to the length of the anchor bolt, and thus the effect of the embedded length on the fatigue behavior of the joint was small. Changes in the slope of the load-displacement curves shown in Fig. 11 can be observed. Unlike

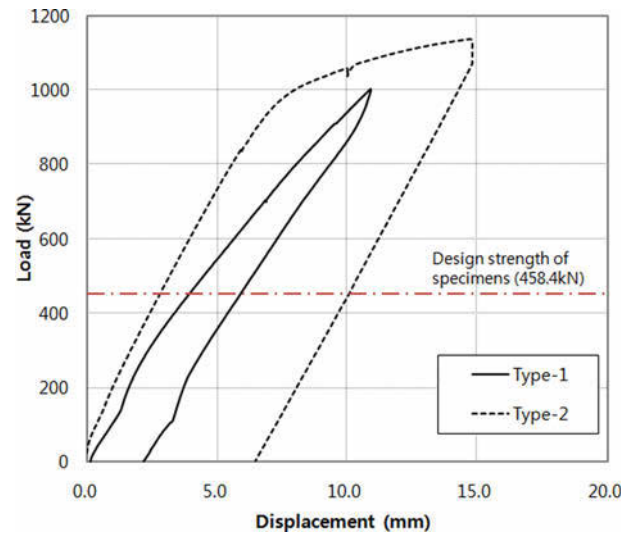


Fig. 11. Load-Displacement Curves for the Specimens under Static Loading

the curve for the Type-1 specimen, which exhibits changes in the slope at initial low loadings, the slope changes for the Type-2 specimen appeared after the pull-out force was greater than 60 kN. In Fig. 11, the slope is large for low loading because there was no detachment between the anchor bolt and concrete; thus, the displacements at the end of the anchor bolt were also small. In contrast, as the pull-out loading increased, the slope decreased because slip occurred at the interface between the anchor bolt and the concrete. The relative displacement at the upper and lower ends of the long anchor bolt (Type-1) was larger than that of the short anchor bolt (Type-2). It was determined that the slip occurred earlier in Type-1 than Type-2, and the slope of the Type-1 curve changed during initial cyclic loading.

After 2 million cycles of cyclic loading were completed, a monotonic and static loading was applied to the specimens. The

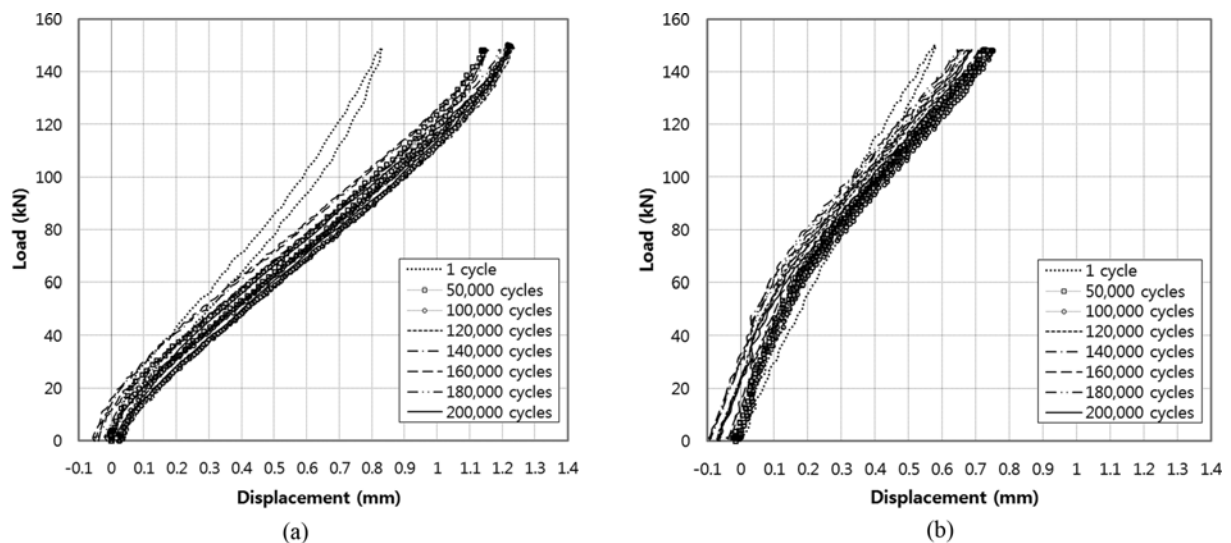


Fig. 10. Load-Displacement Curves for Specimens under Cyclic Loading: (a) Type-1, (b) Type-2

load-displacement curves for each specimen are shown in Fig. 11, which indicates that the test specimens have a yield capacity of 1,000 kN. The load capacity is more than twice the magnitude of the required loading (458.4 kN); thus, it is concluded that the joint has sufficient load capacity after cyclic loading.

The results of the static stress analyses performed to verify the dimension of the specimens were compared with the experimental results; the numerical displacement by 147.1 kN was compared with the experimental results by the first cyclic loading, and the numerical displacement by 1,000 kN was compared with the experimental results by monotonic loading after two million cycles. As a result of the comparison, the displacement and the tendency were similar as shown in Table 9. The only difference was that the experimental displacement due to the monotonic loading was larger than numerical result due to the cyclic loading effect. When the elongations of the anchor bolts were analytically calculated without concrete, the displacements were resulted in 1.5 mm for Type-1 and 1.0 mm for Type-2. These values were 87% and 43% larger than the numerical results, respectively. It meant that the displacement was relatively reduced due to the attachment effect in case of the long anchor bolt, but the displacement was more influenced by the lengths of the anchor bolts than the attachment force.

4.3 Strain

Although the strain gages used in these experiments provided strain histories with insufficient resolution to capture changes in the strain, most of the strain increments during each loading cycle were less than 10 $\mu\text{m/m}$; the strain of the external surface by 147.1 kN was calculated as 7.69 $\mu\text{m/m}$, which was too small to measure it. However, the total strains increased as the number of loading cycles increased. The strain changes from SG1 attached on the Type-1 specimen are shown in Fig. 12, and those from other strain gages, which had similar trends to SG1, are excluded. In the strains listed in Tables 12 to 15, the minus symbol (–) means compression.

Figures 13 and 14 show the accumulated strain histories from each strain gage. The maximum strain of the Type-1 specimen occurred at the lower end of the anchor bolt (SG3), and the maximum strain at the other locations were similar to each other. For the Type-2 specimen, the maximum strain occurred at 45° above the lower end of anchor bolt (SG6). Strains from SG1 and SG3 are excluded in Fig. 14 because SG1 was placed lower than planned and SG3 was damaged during the cyclic loading; the tensile strain appeared at SG1. Regardless of anchor bolt length, the strain increased as the number of load cycles increased, and the overall stiffness of the joint deteriorated.

Furthermore, the strain increment from the first cycle to 1

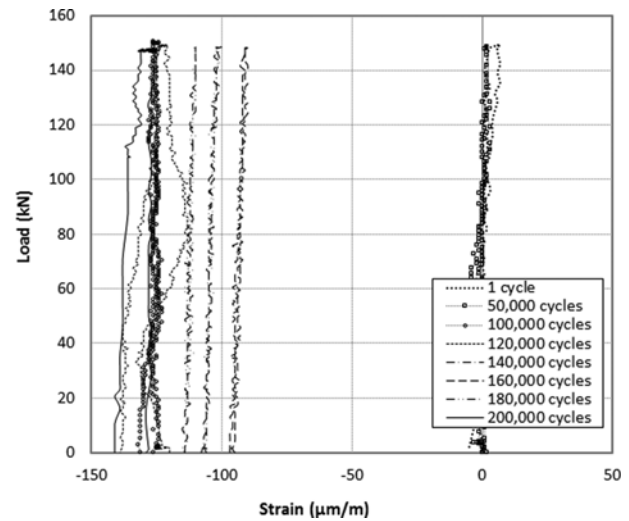


Fig. 12. Load-Strain Curves of a Specimen under Cyclic Loading (Type-1, SG1)

million cycles was large, but was relatively small after 1 million cycles. The reason was expected that the stress distribution ratio between concrete and anchor had changed according to S-N curve changed after 1 million cycles. It also seemed that the load transferred to the concrete was reduced due to the different attachment state between the two materials.

The effect of the embedded length of the anchor bolt was also analyzed using the strain data. For the Type-2 specimen, strains near the embedded part of the anchor bolt were relatively uniform, and this resulted in overall stiffer behavior of the specimen. On the other hand, for the Type-1 specimen which has a longer anchor bolt, differences between the strains at the top end of the anchor bolt and other areas led to more flexural behavior than was observed in Type-1. Thus, it can be concluded that a joint with a longer anchor bolt makes the concrete more flexural.

Figure 15 shows the load-strain curves for the specimens subjected to static loading after completing 2 million cycles of cyclic loading. As the load was statically applied, the strains increased. Because the maximum load was greater than 1,000 kN, residual strains remained at the strain gage points after unloading.

4.4 Cracks

During cyclic loading, only micro-cracks were observed in the specimen until 2 million cycles, which includes small cracks with widths of less than 0.05 mm that occurred after 1.8 million cycles. It seems that cracks were controlled by the reinforcement bars which resist thermal expansion and shrinkage of the concrete, and consequently the specimens behaved as uncracked concrete sections.

Table 9. Comparison of Displacement by Numerical and Experimental Results

Static load	Displacement (Type-1)		Displacement (Type-2)	
	Numerical result	Experimental result	Numerical result	Experimental result
147.1 kN	0.8 mm	0.8 mm	0.7 mm	0.6 mm
1,000 kN	8.7 mm	10.9 mm	5.8 mm	7.9 mm

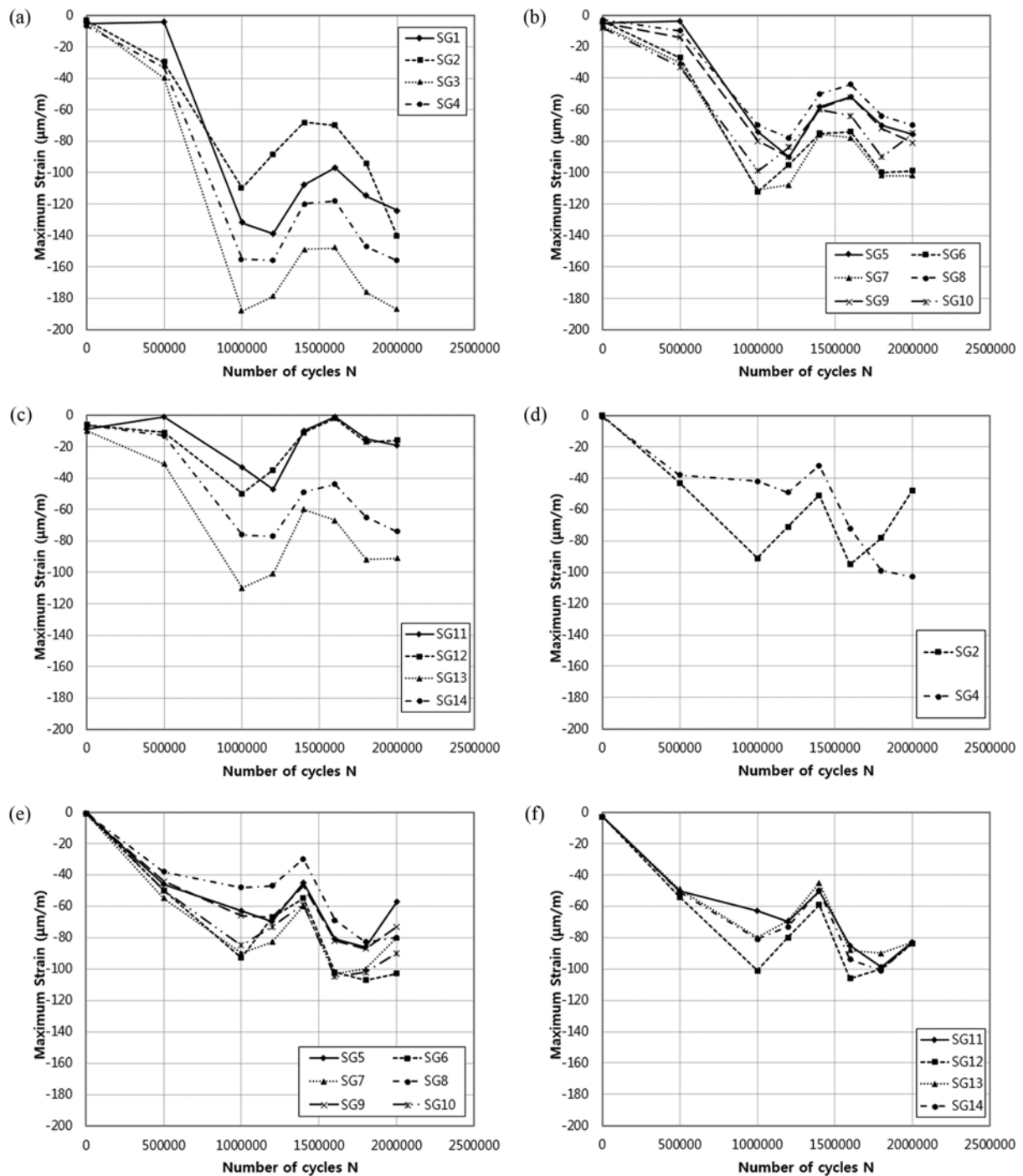


Fig. 13. Change in the Maximum Strain in the Perimetric Direction: (a) Type-1 (1), (b) Type-1 (2), (c) Type-1 (3), (d) Type-2 (1), (e) Type-2 (2), (f) Type-2 (3)

On the other hand, after cyclic loading, severe cracks occurred across the cross-section of the Type-2 specimen during static loading of over 1,000 kN. As shown in Fig. 16, the cracks are located at a line 450 mm below the lower end of the anchor bolt, and its direction is perpendicular to the axis of the anchor bolt. Analyzing the crack pattern suggests that compressive and tensile stresses occurred at the front and the rear of the anchor base plate, respectively.

5. Conclusions

In this study, a fatigue design approach for the joint in hybrid wind turbine towers was proposed as a combination of the conventional design codes. In addition, through several laboratory fatigue tests of a simplified joint specimen, the fatigue performance of the joint was experimentally estimated, and the validity of the proposed design method was demonstrated. The fatigue tests

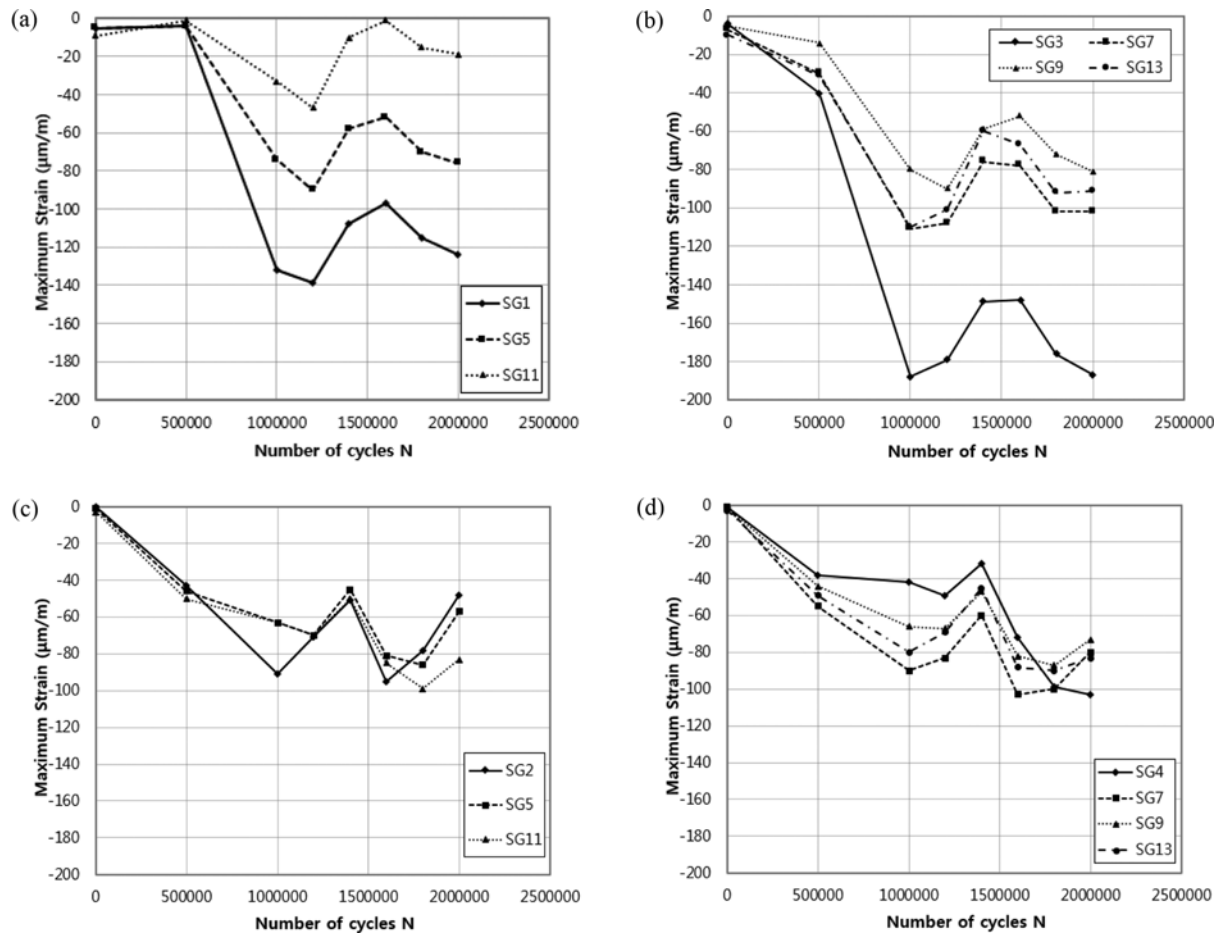


Fig. 14. Change in the Maximum Strain in the Longitudinal Direction: (a) Type-1 (1), (b) Type-1 (2), (c) Type-2 (1), (d) Type-2 (2)

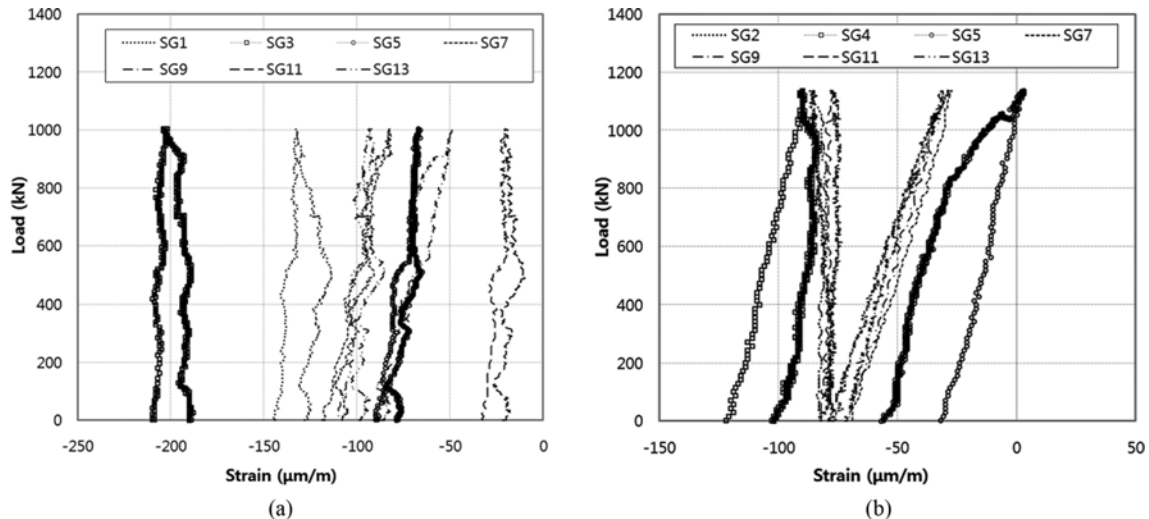


Fig. 15. Load-Strain Curves for the Specimens under Static Loading: (a) Type-1, (b) Type-2

indicated that joints using anchor bolts for hybrid wind turbine towers not only have sufficient fatigue performance, but also safety under static loads. The following are detailed findings drawn from this study:

1. Guidelines in the “CEB-FIP model code 1990” and “Euro-code 2” for fatigue design were adopted and combined to

evaluate the fatigue safety of the joint. As a result, it was shown that the anchor bolt connecting the steel and concrete parts of the tower can resist the equivalent fatigue with 2 million cycles.

2. The joint with anchor bolts shows less decrease in stiffness under cyclic loading. As only micro-cracks with widths of

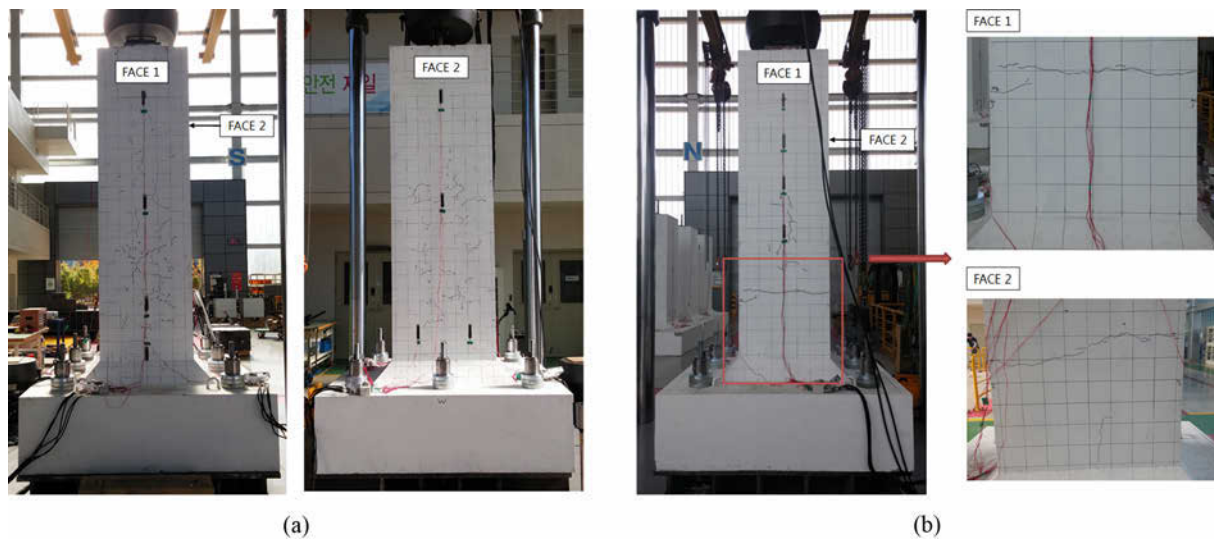


Fig. 16. Crack Distribution after the Static Loading Test: (a) Type-1, (a) Type-2

less than 0.05 mm were observed in the test specimen during cyclic loading of 2 million cycles, the fatigue safety of the joint was also verified. Furthermore, behavior of the Type-1 and Type-2 specimens was similar, and effect of the embedded length was marginal.

3. From the static loading test, a yield capacity of the specimen was estimated to be 1,000 kN, and thus the joint is safe for static loading as well.

Acknowledgements

This study was supported by Korean Institute of Energy Technology Evaluation and Planning (KETEP) granted financial resource from the Ministry of Trade, Industry & Energy, Republic of Korea (No. 20123010020060).

References

- Alvarez-Anton, L., Koob, M., Diaz, J., and Minnert, J. (2016). "Optimization of a hybrid tower for onshore wind turbines by building information modeling and prefabrication techniques." *Visualization in Engineering*, Vol. 4, No. 1, p. 3, DOI: 10.1186/s40327-015-0032-4.
- Bai, X., He, M., Ma, R., Huang, D., and Chen, J. (2017). "Modelling fatigue degradation of the compressive zone of concrete in onshore wind turbine foundation." *Construction and Building Materials*, Vol. 132, pp. 425-437, DOI: 10.1016/j.conbuildmat.2016.12.026.
- CEB (Comite Euro-International du Beton) (1993). *CEB-FIB model code 1990*, Thomas Telford service Ltd., London, England.
- Delhomme, F., Debicki, G., and Chaib, Z. (2010). "Experimental behaviour of anchor bolts under pullout and relaxation tests." *Construction and Building Materials*, Vol. 24, No. 3, pp. 266-274, DOI: 10.1016/j.conbuildmat.2009.08.038.
- Delhomme, F., Thierry, R., Arrieta, B., and Limam, A. (2016). "Pullout behavior of cast-in-place headed and bonded anchors with different embedment depths." *Materials and Structures*, Vol. 49, No. 5, pp. 1843-1859, DOI: 10.1617/s11527-015-0616-4.
- EN 1992-1-1 (2004). *Eurocode 2: Design of concrete structures — Part 1-1: General rules and rules for buildings*, EN 1992-1-1:2004, European Committee for Standardization, Brussel, Belgium.
- EN 1993-1-1 (2005). *Eurocode 3: Design of steel structures — Part 1-1: General rules and rules for buildings*, EN 1993-1-1:2005, European Committee for Standardization, Brussel, Belgium.
- EWEA (The European Wind Energy Association) (2009). *The economics of wind energy*, European Wind Energy Association, Brussels, Belgium.
- GL (Germanischer Lloyd) (2010). *Rules and guidelines IV industrial services. Part 1: Guideline for certification of wind turbines*, Germanischer Lloyd Inderstiel Service GmbH, Hamburg, Germany.
- GWEC (Global Wind Energy Council) (2016). *Global wind report 2016*, <http://gwec.net/publications/global-wind-report-2/global-wind-report-2016/> [Accessed on August 12, 2017].
- Hariyadi, Munemoto, S. and Sonoda, Y. (2017). "Experimental analysis of anchor bolt in concrete under the pull-out loading." *Procedia Engineering*, Vol. 171, pp. 926-933, DOI: 10.1016/j.proeng.2017.01.391.
- KCI (Korea Concrete Institute) (2012). *Korea structural concrete design code and commentary*, Korea Concrete Institute, Seoul, Korea.
- Petersen, C. (1998). "Fatigue assessment eccentrically loaded ring flanges." *Stahlbau*, Vol. 67, No. 3, pp. 191-203.
- Rodriguez, M., Lotze, D., Gross, J. H., Zhang, Y. G., Klingner, R. E., and Graves, H. L. (2001). "Dynamic behavior of tensile anchors to concrete." *ACI Structural Journal*, Vol. 98, No. 4, pp. 511-524.
- Schmidt, H. and Neuper, M. (1997). "Elastic-static behaviour of eccentrically tensioned L-connections with pre-stressed bolts." *Stahlbau*, Vol. 66, No. 3, pp. 163-168.
- Staffan, E., Lynner, T., Hassanzadeh, M., Stalin, T., and Johansson, J. (2010). *Tall tower for large wind turbines*, Report No. 10:48, Swedish Electricity Company Research and Development, Stockholm, Sweden.
- Von der Haar, C. and Marx, S. (2015). "Design aspects of concrete towers for wind turbines." *Journal of the South African Institution of Civil Engineering*, Vol. 57, No. 4, pp. 30-37, DOI: 10.17159/2309-8775/2015/v57n4a4.